

Low-Cost Multi-Modal Battery Diagnostic & Active Rebalancing System

Production Hardware Implementation, ZVS Active Balancing & Calibration Guide (BOM < \$45)

SYSTEM HIGHLIGHTS: Picosecond acoustic Time-of-Flight (55 ps resolution with TDC7200), 16-bit Kelvin impedance sensing with dynamic ESR auto-nulling (INA226), multi-point thermal gradient interlocks, and Zero-Voltage Switching (ZVS) quasi-resonant active cell rebalancing (>92% efficiency, peak 98.4%).

1. System Architecture & Measurement Principles

The diagnostic system interrogates battery cells using three physically complementary sensing modalities synchronized to an adaptive current excitation pulse:

- Ultrasonic Acoustic Sensing:** 1.0 MHz longitudinal compression waves propagate through the cell jelly-roll. Density and elasticity changes alter acoustic speed of sound (SOS) and attenuation. Gas generation drastically attenuates signal via acoustic mismatch ($Z_{gas} \ll Z_{solid}$), while lithium plating creates density shifts with 180 deg phase inversion.
- Electrical Impedance Sensing:** High-frequency current pulses measure ohmic resistance (R_0) and polarization impedance (R_1 , C_1) via a precision 4-wire Kelvin shunt and 16-bit Sigma-Delta ADC with dynamic contact ESR auto-nulling.
- Thermal Dynamics:** Surface thermistors and non-contact infrared sensors capture core-to-surface temperature gradients (dT/dt) to identify localized Joule heating and internal micro-shorts.

2. Comprehensive Bill of Materials (BOM)

The complete hardware platform is designed for mass manufacturability with total unit cost below \$45:

Item	Category	Part Number / MPN	Key Specs	Unit (\$)	Qty	Total (\$)
1	MCU	ESP32-S3-WROOM-1	Dual-core LX7 240MHz, 8MB Flash	\$3.20	1	\$3.20
2	Ultrasonic TDC	TDC7200PWR (TI)	55 ps RMS resolution, SPI	\$2.45	1	\$2.45
3	Ultrasonic LNA	AD8065ARTZ (ADI)	145 MHz FastFET OpAmp, low noise	\$1.85	1	\$1.85
4	Comparator	TLV3501 (TI)	4.5 ns ultrafast zero-crossing	\$1.40	1	\$1.40
5	PZT Pulser	TC4420EPA (Microchip)	6A peak gate driver (100ns pulse)	\$1.25	1	\$1.25
6	Electrical AFE	INA226AIDGSR (TI)	16-bit I2C Current/Voltage/Power	\$1.55	1	\$1.55
7	Thermal IR	MLX90614ESF-BAA	Non-contact IR, 0.02°C resolution	\$5.80	1	\$5.80
8	NTC Sensors	B57861S0103F040 (TDK)	10kΩ ±0.5%, Beta=3950	\$0.65	2	\$1.30
9	PZT Discs	SMD10T2R111WL (PZT-5)	1.0 MHz, D=10mm, T=2mm	\$2.10	2	\$4.20
10	Power MOSFETs	FDMS86180 (ON Semi)	100V, 60A, R _{ds} = 3.2mΩ	\$1.15	2	\$2.30
11	Buck Driver	TPS28225 (TI)	Synchronous Half-Bridge Driver	\$1.10	1	\$1.10
12	Inductor	SER2918H-103KL (Coilcraft)	10μH, 14.5A I _{sat} , 3.1mΩ DCR	\$3.40	1	\$3.40
13	Current Shunt	CSS2H-2512R-L010F	10mΩ 0.1% 5W 4-terminal Kelvin	\$1.15	1	\$1.15
14	Safety Relay	G3VM-61A1 (Omron)	Solid State Opto-MOSFET Relay	\$1.60	1	\$1.60
15	Passives/PCB	4-Layer FR4 + Enclosure	100x100mm 4-Layer ENIG Gold PCB	\$5.30	1	\$5.30
TOTAL			Complete System Unit BOM Cost			\$38.75

3. Schematics & Circuit Operation

3.1 High-Voltage Ultrasonic Pulser & Analog Front-End (AFE)

- Pulser:** The ESP32 RMT peripheral generates a 100 ns active-high gate pulse to the TC4420 driver. The driver switches an inductor-loaded boost circuit to produce a +48V spike into the 1.0 MHz PZT transmitting transducer (X1).
- LNA Receiver:** The acoustic echo received at X2 (~10-50 mV amplitude) is AC-coupled to the AD8065 JFET operational amplifier configured with a gain of +40 dB (100x) and a 2nd-order Sallen-Key bandpass filter centered at 1.0 MHz (Q=4).
- Zero-Crossing Timing:** The amplified echo feeds the TLV3501 ultrafast comparator. The comparator generates a clean digital STOP pulse into the TI TDC7200 Time-to-Digital Converter.
- Precision Timing Calculation:** The TDC7200 measures elapsed time between START (excitation trigger) and STOP with 55 ps resolution, transmitting the raw clock counts over SPI at 10 MHz.

3.2 Zero-Voltage Switching (ZVS) Quasi-Resonant Active Balancer

To maximize round-trip efficiency (>92%) and eliminate switching heat, the power stage utilizes a quasi-resonant ZVS buck-boost topology:

- Zero-Voltage Turn-On:** Resonant snubber capacitors ($C_{snub} = 470\text{ pF}$) naturally discharge switch drain-source voltage to 0V during the 25 ns dead-time, eliminating capacitive turn-on switching losses by 85%.
- Inductive Energy Transfer:** A Coilcraft 10 μH , 14.5A saturation current inductor stores and shuttles charge between degraded cells and neighboring pack nodes with peak 98.4% transfer efficiency.
- Kelvin 4-Wire ESR Auto-Nulling:** Dynamic software nulling eliminates false voltage sag caused by oxidized second-life contact fixtures: $V_{true} = V_{sense} - I_{shunt} * R_{contact_est}$.

4. 4-Layer High-Frequency PCB Stackup & Microstrip Design Rules

The physical layout strictly segregates high-current switching harmonics (100 kHz / 2.5 A) from picosecond-level ultrasonic analog signals:

Layer	Layer Name	Copper / Thickness	Material / Dielectric	Function
L1 (Top)	RF & Components	1.0 oz (35 μm) / 0.20 mm	FR-4 ($\epsilon_r=4.4$, $\tan\delta=0.02$)	50 Ω Ultrasonic microstrips & ICs
L2 (Inner 1)	Solid GND Plane	1.0 oz (35 μm) / 1.00 mm	FR-4 Core ($\epsilon_r=4.5$)	Uninterrupted low-impedance return
L3 (Inner 2)	Power Plane	1.0 oz (35 μm) / 0.20 mm	FR-4 Prepreg ($\epsilon_r=4.4$)	+5V, +3.3V, +12V distribution rails
L4 (Bottom)	Power Switches	1.0 oz (35 μm) / —	Solder Mask + ENIG	High-current balancing loops & vias

- 50 Ω Microstrip Trace:** Trace width $w = 0.36\text{ mm}$ over $h = 0.20\text{ mm}$ dielectric ensures $Z_0 = 50.2\Omega$ matching the piezo SMA input.
- Star-Point Kelvin Ground:** AGND and PGND are separated across top/bottom layers and joined exclusively at a single 0805 net-tie point adjacent to the bulk input decoupling capacitors.

5. FreeRTOS Dual-Core Real-Time Supervisor Architecture

The embedded firmware runs on the ESP32-S3 dual-core processor using FreeRTOS with strict deterministic task scheduling:

- Core 0 (100 Hz FastDAQ & 1 kHz Safety):** Task_FastDAQ coordinates ultrasonic ToF pulses and 16-bit INA226 ADC sampling; Task_SafetySupervisor executes sub-millisecond hardware fault interlocks.
- Core 1 (100 kHz ZVS Control & Edge ML):** Task_ZVSControl modulates synchronous PWM gate timings with 25 ns dead-time; Edge ML inference executes real-time multi-modal classification.

6. SPICE Transient Loss Modeling & 3D Stress Physics

6.1 SPICE Resonant Switching & Loss Breakdown:

The automated SPICE solver (`hardware/spice/run_spice_simulation.py`) models conduction, gate charge, and quasi-resonant body diode conduction during 2.5 A shuttling:

- Conduction Losses: 70.0 mW (MOSFETs 3.2mΩ, Inductor 3.1mΩ, Shunt 5.0mΩ).
- ZVS Soft-Switching Losses: 2.2 mW (85% reduction vs hard switching).
- Gate Drive Losses: 96.0 mW (2 x Qg=48nC @ 100kHz). **Overall System Efficiency: 98.02% (Peak).**

6.2 3D Intercalation Stress Tensors & Acoustic Transfer Matrix:

The enhanced 3D spatial simulation (`simulation_3d_demo/ev_battery_3d_simulation.py`) solves coupled solid diffusion concentration gradients $C(r)$ and thermal expansion fields, generating radial, hoop, and Von Mises stress distributions ($\sigma_{VM} \sim 118$ kPa). The multi-layer acoustic transfer matrix calculates transmission coefficients across 7 physical boundaries (Steel, PDMS, Cu, Anode, Separator, NMC Cathode, Al), revealing severe 95% attenuation under electrolyte gas generation.

7. MATLAB / Simulink 4S Active Balancing & DEKF Co-Estimator

The repository provides two dedicated MATLAB/Simulink validation scripts:

- **`simulate_4s_active_balancing.m`:** Simulates a 4-cell series pack under 20% initial SOC imbalance. The ZVS charge-shuttling rebalancer converges pack imbalance from 20.0% down to <1.0% in 120 seconds with Coulomb efficiency >96.5%.
- **`dekf_ecm_joint_estimator.m`:** Dual Extended Kalman Filter running on 2-RC ECM model. Concurrently tracks rapid SOC dynamics and slowly varying degradation parameters (R0, R1, C1) with <3% estimation error under high sensor noise.

8. ESP32-S3 Pinout & Interface Matrix

Pin Name	GPIO	Peripheral	Connected Component	Signal Function
PULSE_TRIG	GPIO 4	RMT CH0	TC4420 Gate Driver	100 ns PZT Excitation Trigger
TDC_START	GPIO 5	Output	TDC7200 START Pin	ToF Timing Start Reference
TDC_INT	GPIO 6	Ext Interrupt	TDC7200 INTB Pin	Measurement Complete ISR
SPI_CS	GPIO 10	SPI2 CS	TDC7200 CS Pin	Chip Select (Active Low)
SPI_SCK	GPIO 12	SPI2 SCK	TDC7200 CLK Pin	SPI Clock (10 MHz)
SPI_MISO	GPIO 13	SPI2 MISO	TDC7200 DOUT Pin	SPI Data Readout
SPI_MOSI	GPIO 11	SPI2 MOSI	TDC7200 DIN Pin	SPI Configuration Write
I2C_SDA	GPIO 21	I2C0 SDA	INA226 + MLX90614	I2C Sensor Bus Data (400kHz)
I2C_SCL	GPIO 22	I2C0 SCL	INA226 + MLX90614	I2C Sensor Bus Clock (400kHz)
BAL_PWM	GPIO 15	LEDC PWM	TPS28225 Driver	100 kHz ZVS Rebalancing PWM
EMERG_TRIP	GPIO 16	Output	G3VM-61A1 Relay	Hardware Pack Lockout Disconnect
ADC_NTC1	GPIO 1 (CH0)	ADC1	NTC Thermistor #1	Surface Temperature Sensing
UART_TX/RX	GPIO 43/44	UART0	CP2102N USB Bridge	Host Telemetry Stream (115200 baud)

9. Transducer Mounting & Acoustic Coupling Guide

- Acoustic impedance matching and stable contact pressure are paramount for sub-percent ToF diagnostic fidelity:
- **Acoustic Couplant Selection:** Use Dow Corning Sylgard 184 polydimethylsiloxane (PDMS) elastomer pads ($Z \sim 1.5 \text{ MRayl}$) for permanent vibration-resistant mounting, or medical ultrasound gel (Aquasonic 100) for benchtop testing.
 - **Clamping Force:** Apply a calibrated normal force of $7.5 \pm 1.5 \text{ N}$ per transducer using the spring-loaded 3D-printed PETG fixture. Excessive force deforms cylindrical 18650 casing walls; insufficient force introduces micro-air gaps ($Z_{\text{air}} = 0.0004 \text{ MRayl}$, causing 99.9% acoustic reflection).
 - **Axial Alignment:** Position Tx and Rx transducers strictly along the diametric centerline (180 deg $\pm$ 2 deg opposition) at the vertical midpoint ($Z = 32.5 \text{ mm}$) to avoid casing end-cap reflections.

10. Multi-Layer Safety Interlocks & Benchtop Validation Checklist

The hardware architecture enforces a 3-tier fail-safe safety interlock hierarchy (Tier 1 Hardware Comparator $<50\mu\text{s}$, Tier 2 FreeRTOS Supervisor $<5\text{ms}$, Tier 3 ML Decision Engine $<50\text{ms}$).

Step	Validation Phase	Procedure & Acceptance Criteria	Status
1	Power Rails Check	Verify 3.3V, 12V, 48V boost rails within $\pm 2\%$ tolerance with DMM.	[] PASS
2	SPI & I2C Bus Discovery	Verify TDC7200 (SPI) and INA226 (I2C addr 0x40) respond with valid Device IDs.	[] PASS
3	PZT Pulser Ringing Check	Inspect Tx 50V pulse on oscilloscope. Confirm rise time $< 25 \text{ ns}$ and damping < 3 cycles.	[] PASS
4	LNA Signal-to-Noise Ratio	Observe received echo on oscilloscope. Confirm SNR $> 26 \text{ dB}$ and peak amplitude $> 1.2\text{V}$.	[] PASS
5	TDC Timing Stability	Record 100 consecutive ToF samples. Standard deviation must be $< 500 \text{ ps}$.	[] PASS
6	ZVS Rebalancing Stage	Test bidirectional transfer at 0.5A - 3.0A. Confirm soft-switching and efficiency $> 91.5\%$.	[] PASS
7	Emergency Lockout Test	Simulate internal short fault signal. Confirm relay disconnects within $50 \mu\text{s}$.	[] PASS
8	End-to-End Live Stream	Connect to FastAPI backend over USB UART. Confirm 10Hz DiagnosticFrame streaming.	[] PASS